

Preliminary Design Presentation

Communication & Charge Portable System



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ELEN 482 - 95

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Name & Logo of the company



- **Eco** is defined as the environment.
- **Circuit** is an interconnection of electric components
- **Leaf** represents the nature.
- **Terminals** represent an electric circuit.

Abstract

- After all the disasters that Hurricane Maria brought in Puerto Rico, we realized how powerful nature can be and how vulnerable we are.
- This disaster brought many consequences to society such as the lack of electricity, also destroyed communication towers.
- After this problem, the Ecocircuit company found a solution to this problem, designed a portable communication and charging system powered by solar panels.
- The output of the project or the mayor accomplishment that Ecocircuit acquired is that they could make it possible to create this system; allowing them to power a communications system, among other systems simultaneously.

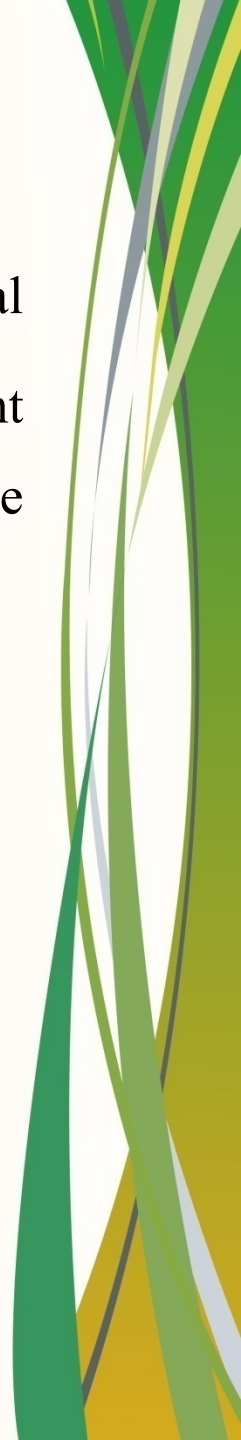


Mission

As a company is to be leaders in self-sustaining projects that use natural resources and the field of electrical engineering to help the environment directly or indirectly. We seek through products to help humanity and the environment

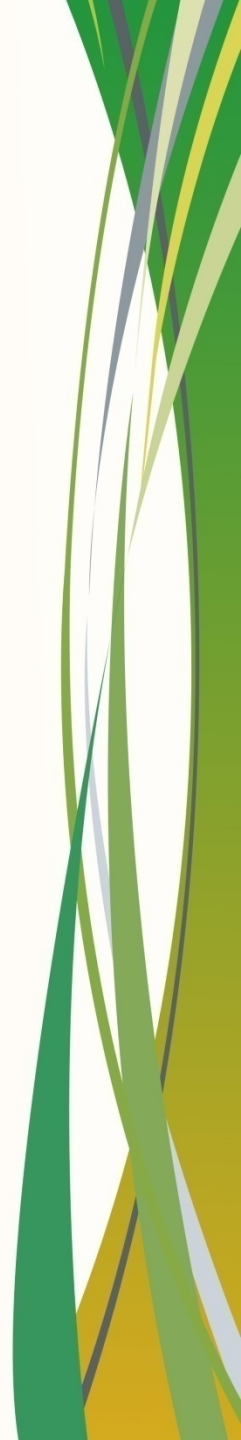
Vision

Develop a variety of devices that contribute to the conservation of the environment that can be used by present and future generations



Problem Description

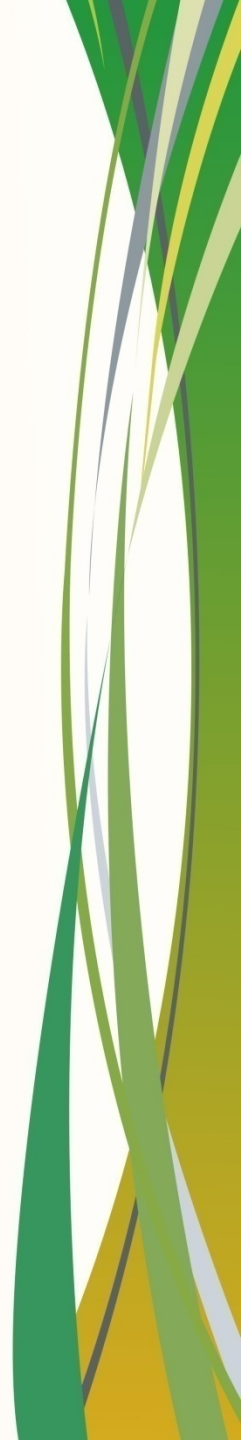
- September 26, 95% no electricity & 95% no cell phone service.
- Two weeks after, 89% still had no electricity, 44% had no water service and 58% had no cell phone service.
- A month after the hurricane, 88% had no electricity, 29% needed running water and 40% of the island had no mobile services.



Solutions

Create a system that:

- Be portable.
- It is energized with sunlight.
- Connects to a satellite internet.
- Have USB ports to charge cell phones and tablets.
- Connect a refrigerator or other small AC devices.



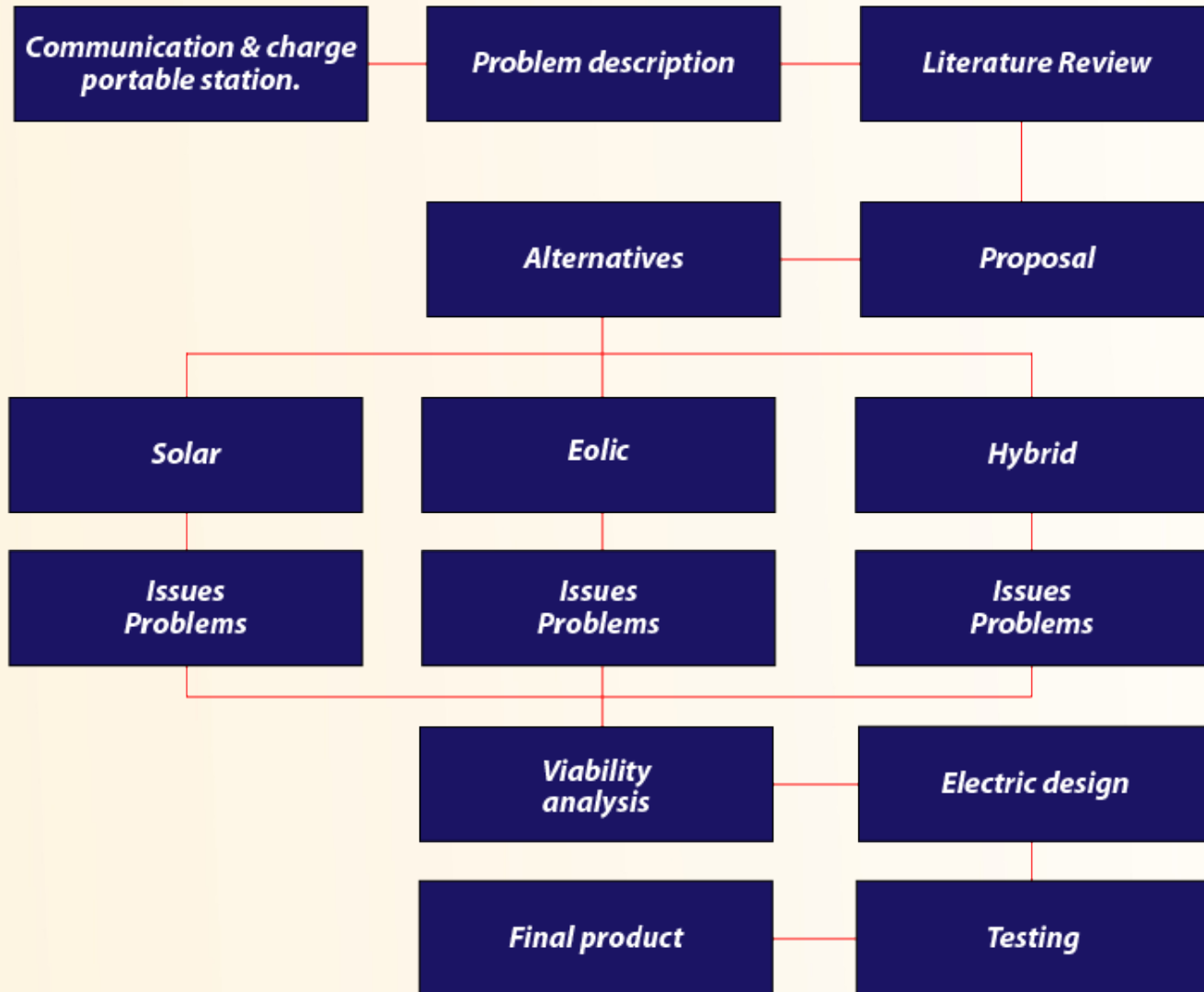
Literature review

- Charges 6 phones
- Portable
- Eco Friendly
- Cost: \$500

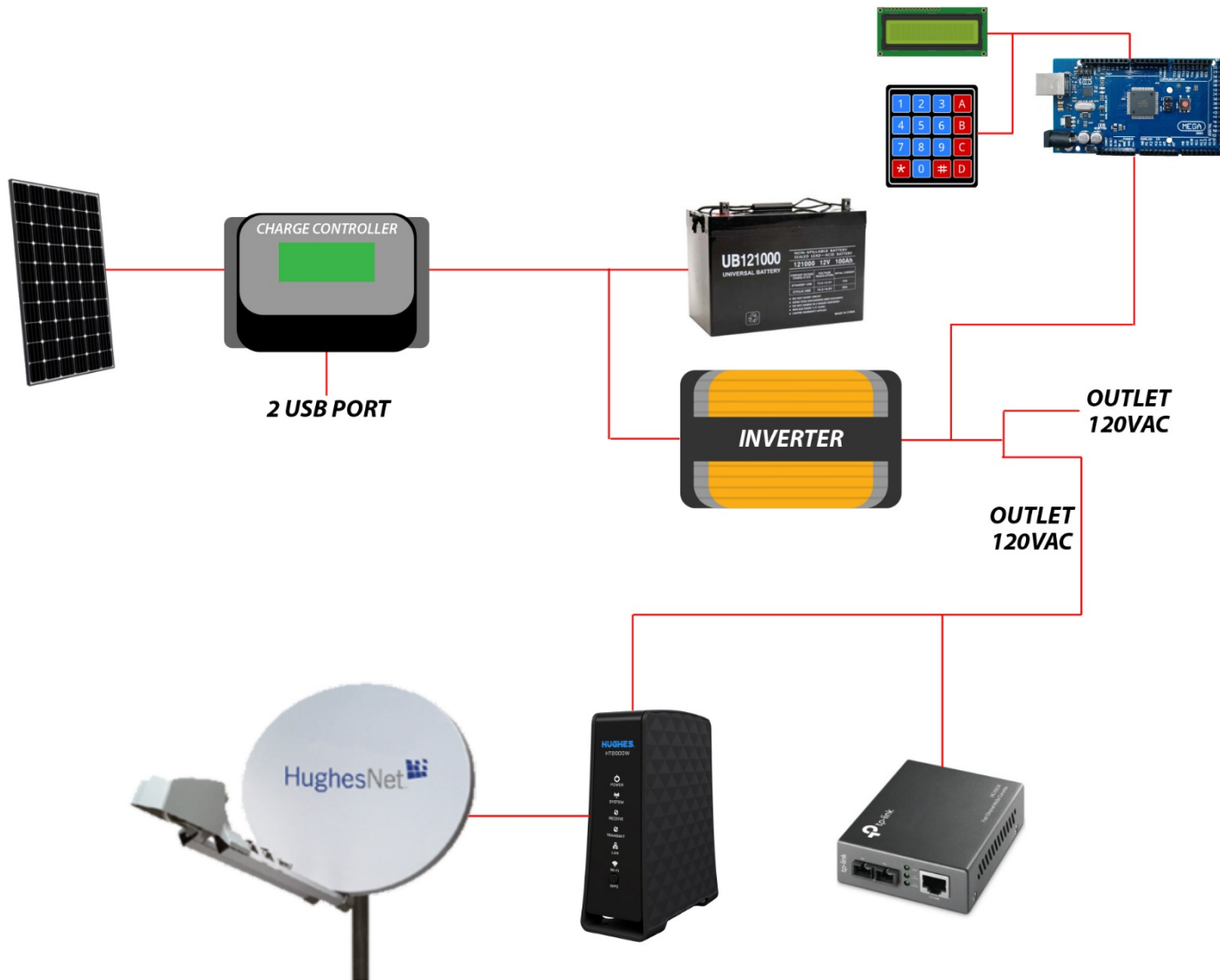


Figure 1: Solar charge station by ©
2015 SUNSUPPLY

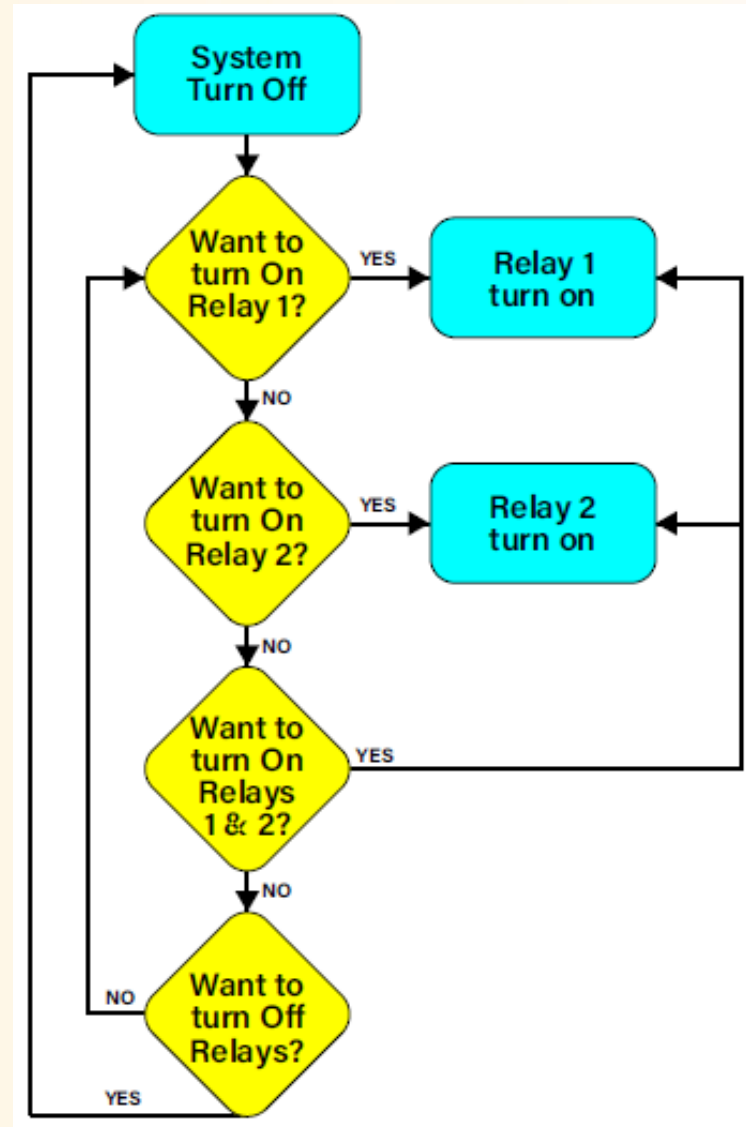
Methodology Diagram



Components System Diagram



Flowchart Diagram



Communication & Charge Portable System



Figure 2: 3D design right view



Figure 3: 3D design left view

Communication & Charge Portable System

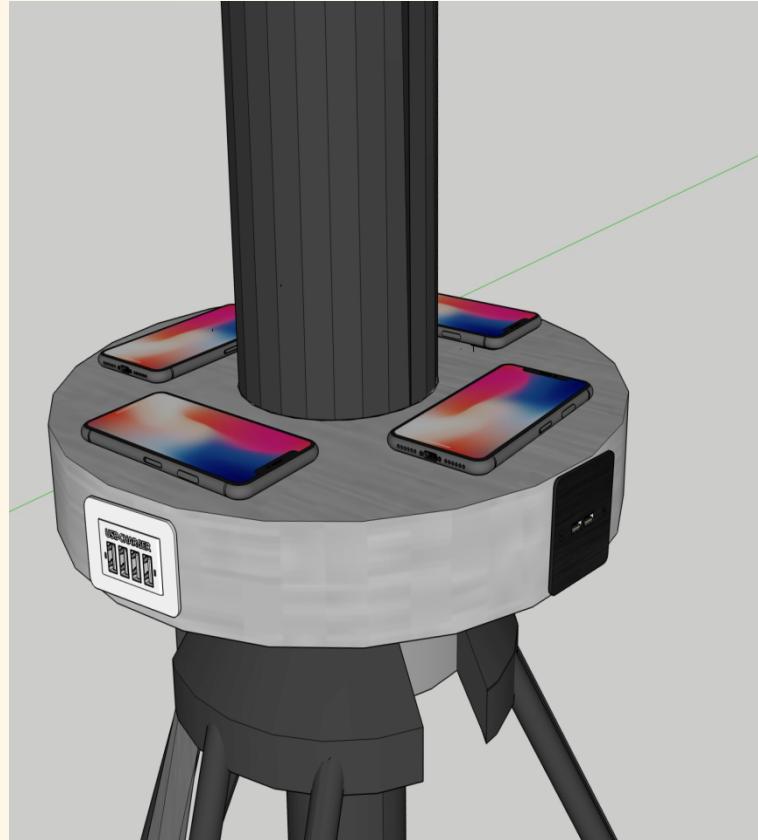


Figure 4: 3D design USB port view

Issues

Communication & Charge Portable System

- **Social**

Helps the community

- **Enviromental**

Eco friendly

- **Economic**

Approx. \$2,107.5

- **Ethical**

Canons of ethics: 1, 4, 5, 10, 12, 14, 17, 18, 19 & 20.

- **Health & Security**

The product is extremely safe. Does not cause diseases to any living being.

- **Construction**

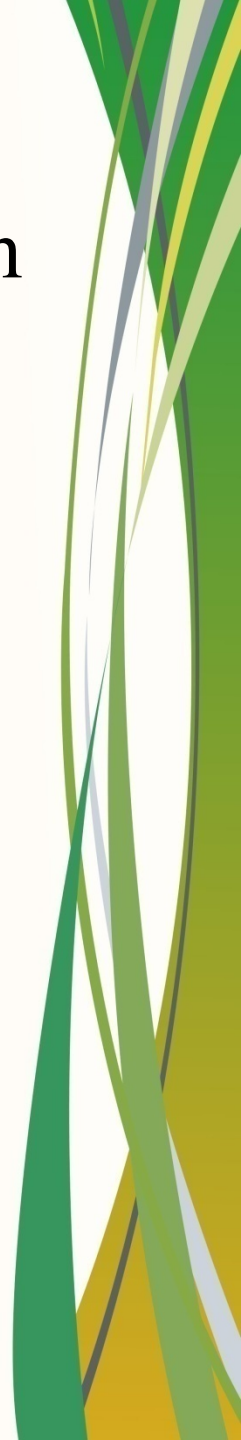
It could be a source of employment.

- **Sustainability**

Requires natural resources such as sun and wind.

- **Problems & Engineering Solutions**

If factory defects occurs the company will send a technician.



Cost Analysis

Components	Cost	
	Design	Reproduction
Fiber Optic	\$ 77.00	\$ 48.00
Solar Tarcking (2)	\$ 242.00	\$ 195.00
Battery	\$ 350.00	\$ 275.00
PV Solar (2)	\$ 612.00	\$ 350.00
Inverter	\$ 70.00	\$ 45.00
Charge Controller	\$ 30.00	\$ 24.00
Arduino Mega	\$ 12.00	\$ 4.00
LCD display	\$ 8.00	\$ 1.00
Keyboard	\$ 5.00	\$ 1.00
Frame	\$ 250.00	\$ 100.00
Installation		\$ 60.00
+25%		\$ 275.75
Total	\$ 1,656.00	\$ 1,378.75

Components	Cost
Inverter \$	49.97
PV Solar \$	11.47
Battery \$	100.34
Outlet \$	1.31
Relays (2) \$	5.28
Arduino Mega \$	7.59
LCD Keypad \$	9.88
Cables \$	12.00
Structure \$	55.12
Arduino UNO \$	4.50
Light Sensor \$	2.09
Stepper Motor \$	8.00
Motor Drive \$	1.79
Total \$	269.34

Table 1. Fiber optic media converter comparison.

Fiber Optic Media Converter			
Name	Ymiko 2PCS SC10/100/1000M	Gigabit Ethernet Media Converter	TP-Link MC100CS
Price (\$)	77.46 (2)	70.99 (2)	44.99 (1)
Extends fiber distance (km)	2	2	2
Dimensions	5.5'' x 4.3'' x 1.2''	93 x 71 x 27 mm	3.7 ''x 2.9'' x 1.1''
Weight (g)	734	500	603
Power consumption	5VDC / 1A	5VDC / 1A	5VDC / 1A
Max Power Consumption (W)	2.5	3	2.5

Table 2. Solar tracking system comparison.

Solar tracking system			
Price (\$)	121.19	154.74	103.89
Nominal output (V)	12-24	12	12
Max load current (A)	8	3	3
No load current (A)	.8	.5	.8
Linear actuator			
Stroke length (in.)	18"	18"	18"
Retracted length (mm)	600	600	600
Extended length (mm)	1050	1050	1050

Table 3. Battery comparison.

Batteries		
Name	Lead Acid battery	Lithium Ion Battery
Capacity (Ah)	200	100
Price (\$)	350	750
Weight (lb.)	72	31
Dimensions (in.)	13.5 x 6.7 x 8.3	12.75 x 6.8 x 9
Efficiency (%)	80	95

Table 4. Solar panels comparison.

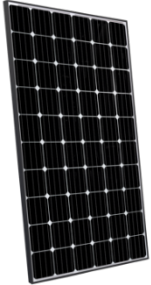
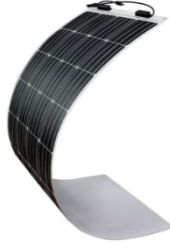
Solar Panels			
Name	Monocrystalline	Polycrystalline	Thin film flexible
Voltage (V)	24	24	18
Watts (W)	300	270	300
Price (\$)	306	225	125
Weight (kg)	19	19	2
Dimensions (in.)	64.2 x 39.2 x 1.5	64.2 x 39.2 x 1.5	45.9 x 21.9 x 0.8
Efficiency (%)	15 - 20	13 - 16	14 - 18

Table 5. Inverter comparison.

Power Inverter			
Name	Ampeak	VertaMax	Potek
Voltage (V)	12	12	12
Watts (W)	1000	1000	2000
Price (\$)	70	100	140
Weight (lb.)	8.8	11.4	10.6
Dimension (in.)	10.3 x 8.3 x 3.4	16.7 x 10.6 x 5.1	16.7 x 7.1 x 12.8

Table 6. Charge controller comparison.

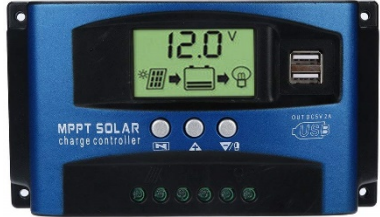
Charge Controller			
Name	EPEVER MPPT Charge Controller	Outback Solar Charge Controller	Sunsee Solar Charger Controller
Battery Voltage (V)	12-24	12-60	12-24
Current Output (A)	30	80	100
Dimensions (in.)	9 x 6.5 x 2.2	16.2 x 5.8 x 4.5	6 x 4 x 14.9
Weight (lb.)	3.53	12.2	11.3
Price (\$)	120	455	24.95

Table 7. Microcontroller comparison.

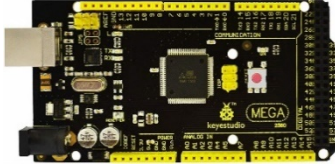
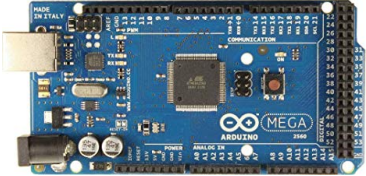
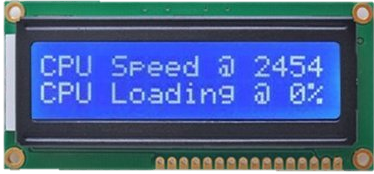
Microcontroller			
Name	KEYESTUDIO Mega 2560 R3 Board	INSIGNIA LABS - ATMEL	ELEGOO MEGA 2560
Voltage (V)	7-12	12	7-12
Dimensions (in.)	4.4 x 1.3 x 2.7	n/a	3.2 x 2.4 x 0.4
Weight (ounce)	2.88	5.3	0.96
Price (\$)	15	19.95	12

Table 8. LCD display comparison.

Types of LCD			
Voltage (V)	5	5	5
Current (A)	0.52	0.67	0.13
Characters	16 x 2	160 x 128	20 x 4
Dimensions (in.)	6 x 2 x 0.3	3 x 2 x 1	3.86 x 2.36 x 0.5
Price (\$)	7.99	28.58	7.99

Formulas and Standards

- Average daily consumption (Wh):

$$Lmd = \frac{Lmd, DC + \frac{Lmd, AC}{N_{inv}}}{N_{bat} * N_{con}}$$

- Average daily consumption (Ah):

$$Q_{ah} = \frac{Lmd}{V_{bat}}$$

- Number of solar panels:

$$N_t = \frac{Lmd}{P_{mpp} * HPS * PR}$$

- Number of solar panels serie:

$$N_{serie} = \frac{V_{bat}}{V_{mod, mpp}}$$

- Number of solar panels parallel:

$$N_{paralelo} = \frac{N_t}{N_{serie}}$$

- Current generated:

$$I_{gfv, mpp} = \frac{Q_{ah}}{HPS}$$

- Battery capacity (Cnd):

$$Cnd (Wh) = \frac{Lmd}{P_{dmax, d} * Fct}$$

$$Cnd (Ah) = \frac{Cnd (Wh)}{V_{bat}}$$

- Battery capacity (Cne):

$$Cne (Wh) = \frac{Lmd * N}{P_{dmax, e} * Fct}$$

$$Cne (Ah) = \frac{Cne (Wh)}{V_{bat}}$$

Formulas and Standards

- **Charge controller current**

$$I_{entrada} = 1.25 * I_{mod,sc} * N_p$$

$$I_{salida} = \frac{1.25 * (P_{dc} + \frac{P_{ac}}{N_{inv}})}{V_{bat}}$$

- **Inverter watts**

$$P_{inv} = 1.2 * P_{ac}$$

- **Ohms law:**

$$V = I * R$$

- **Watts:**

$$W = V * I$$

- **Sun in Puerto Rico**

$$HSP = 5.53 \text{KW/m}^2$$

- **NFPA 70 National Electric Code**

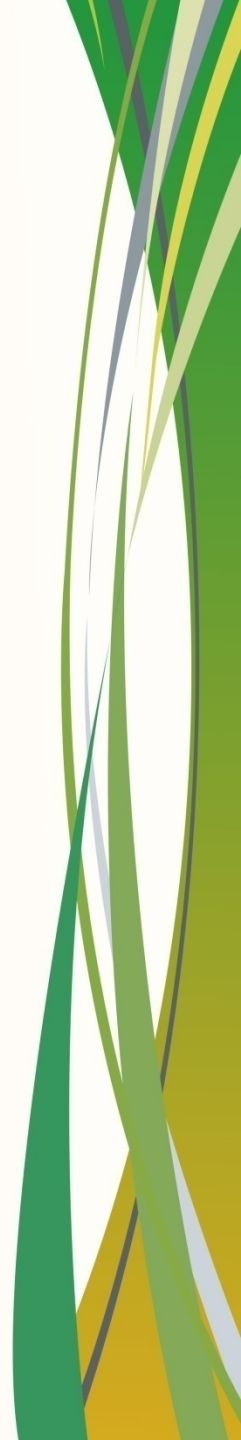
Article 110: Requirements
of electrical installation

Article 250: Grounding

Article 645: Electronic
processing equipment

Article 770: Cable

- **NFPA 101 Life Safety Code**



Prototype



Figure 5: prototype construction 1



Figure 6: prototype construction 2

Prototype

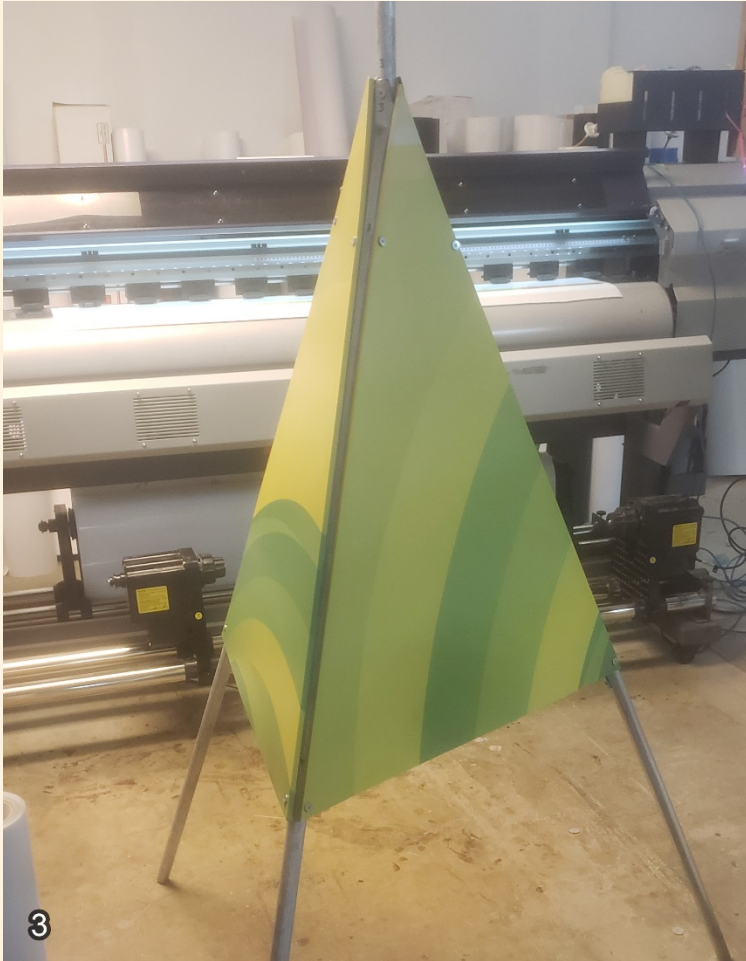


Figure 7: prototype construction 3



Figure 8: prototype construction 4

Prototype



Figure 9: prototype construction 5



Figure 10: identified prototype 1

Prototype

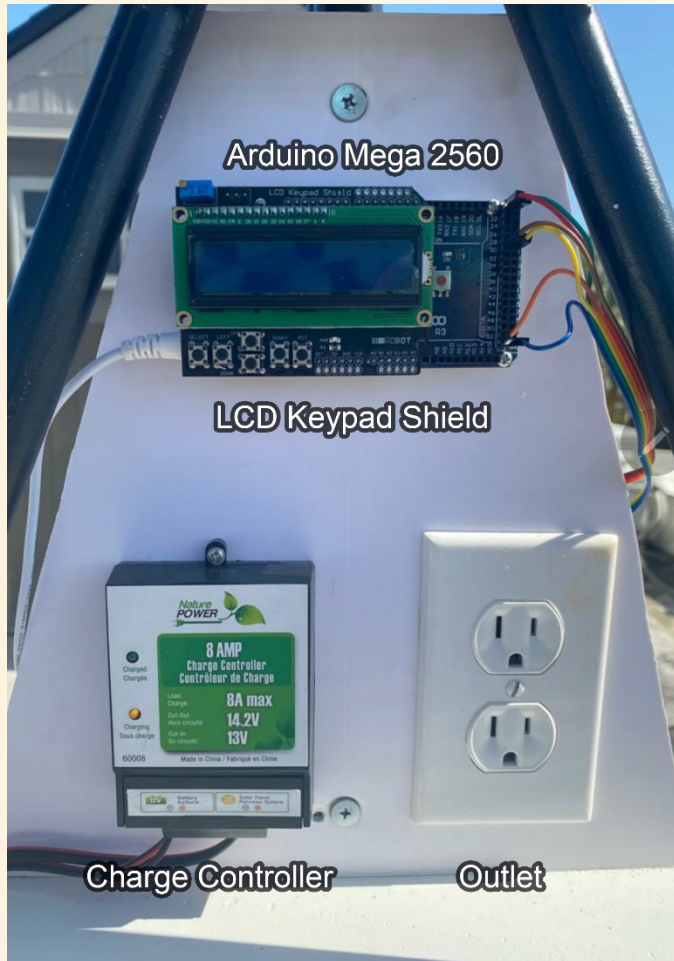


Figure 11: identified prototype 2



Figure 12: identified prototype 3

Prototype specs

- Dimensions (L x W x H) = 41"x41"x77"
- Weight :
 - Battery = 22.8 lbs.
 - Sola Panel = 5.1 lbs.
 - **Total** = 56.2 lbs.
- Cost :
 - Inverter 300W = \$49.97
 - Solar Panel 22W & Charge Controller 8A = \$111.47
 - Battery 35Ah = \$100.34
 - Outlet = \$1.31
 - 2 Relays = \$5.28
 - Arduino Mega 2560 = \$7.59
 - LCD Keypad Shield = \$9.88
 - Cables = \$12.00
 - Structure = 55.12
 - PVC 1/2" = \$37.00
 - Metal 3/4" = \$18.12
 - Arduino UNO = \$4.50
 - 2 Light sensor = \$2.09
 - Stepper Motor = \$8.00
 - Stepper Motor Driver = \$1.79
 - **Total** = \$369.34

Prototype specs

- Measured :
 - Current Consumption:
 - Arduino Mega = .093A
 - Arduino UNO = .046A
 - Relay = .16A
 - Stepper Motor = 1.65A
 - ARRIS TG862A Modem = .48A
 - Linksys E1000 Router = .5A
 - **Total** = 2.929A
- Time Response :
 - Arduino Mega 2560 = 2s
 - Arduino UNO = 2s
 - Inverter = 500 ms
 - Relay = 500 ms

Stepper motor



Figure 13: stepper motor construction 1



Figure 14: stepper motor construction 2

Battery Voltage Reading



Figure 15: LCD screen battery voltage reading reduction

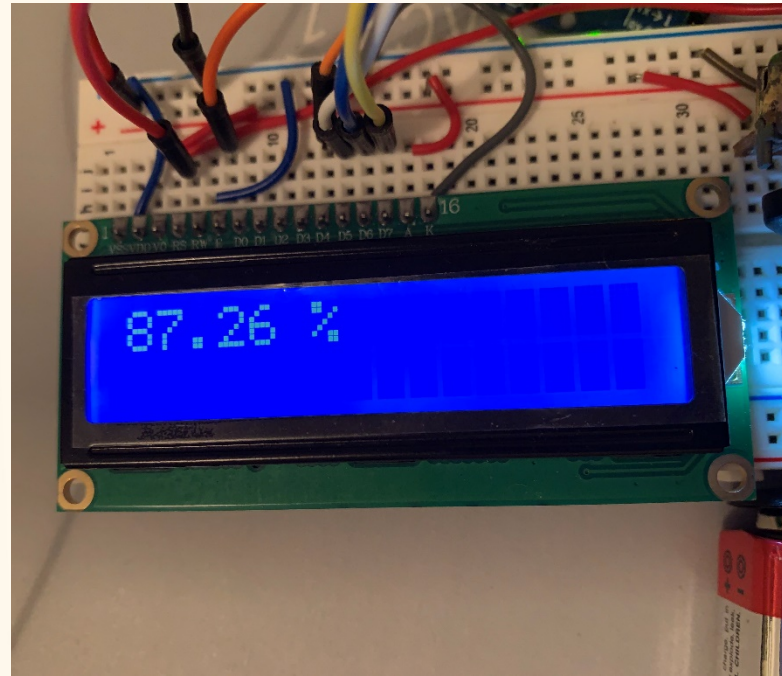
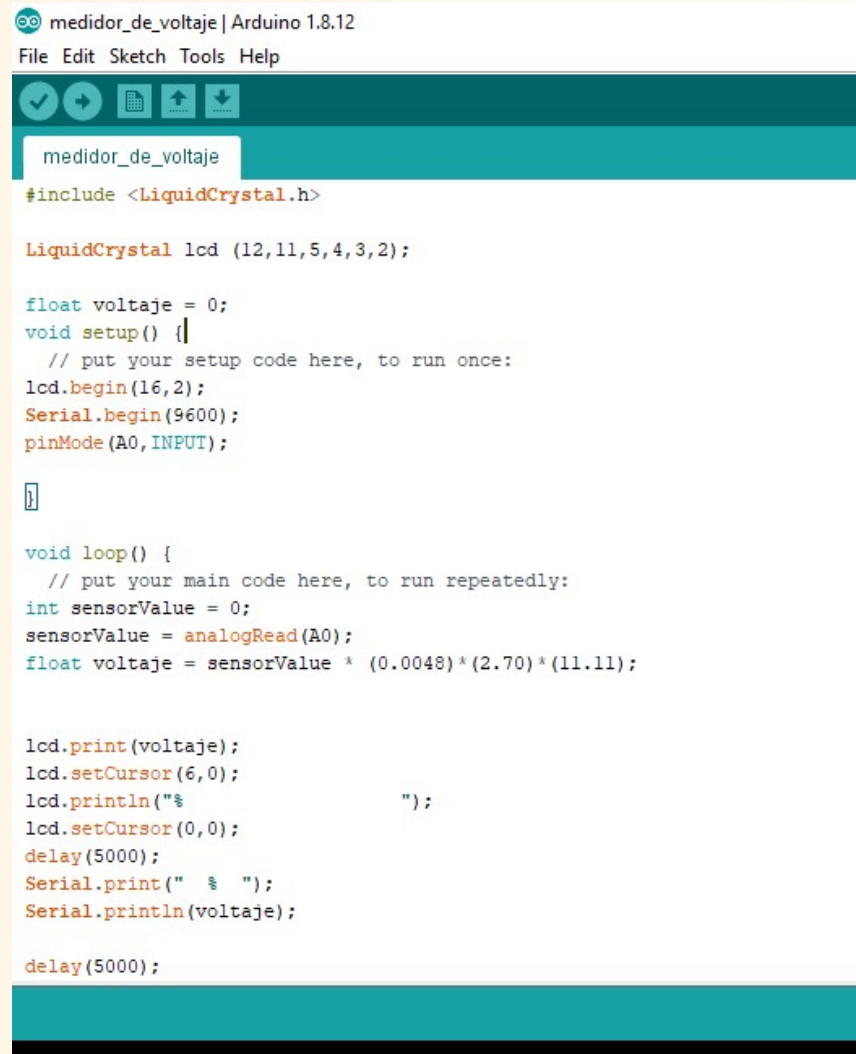


Figure 16: LCD screen battery voltage reading reduction

Battery Voltage Reading

The image shows a screenshot of the Arduino IDE interface. At the top, the title bar reads 'medidor_de_voltaje | Arduino 1.8.12'. Below it is a menu bar with 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. A toolbar with icons for checking, running, serial monitor, and file operations is visible. The main text area contains the following code:

```
medidor_de_voltaje
#include <LiquidCrystal.h>

LiquidCrystal lcd (12,11,5,4,3,2);

float voltaje = 0;
void setup() {
  // put your setup code here, to run once:
  lcd.begin(16,2);
  Serial.begin(9600);
  pinMode(A0,INPUT);

}

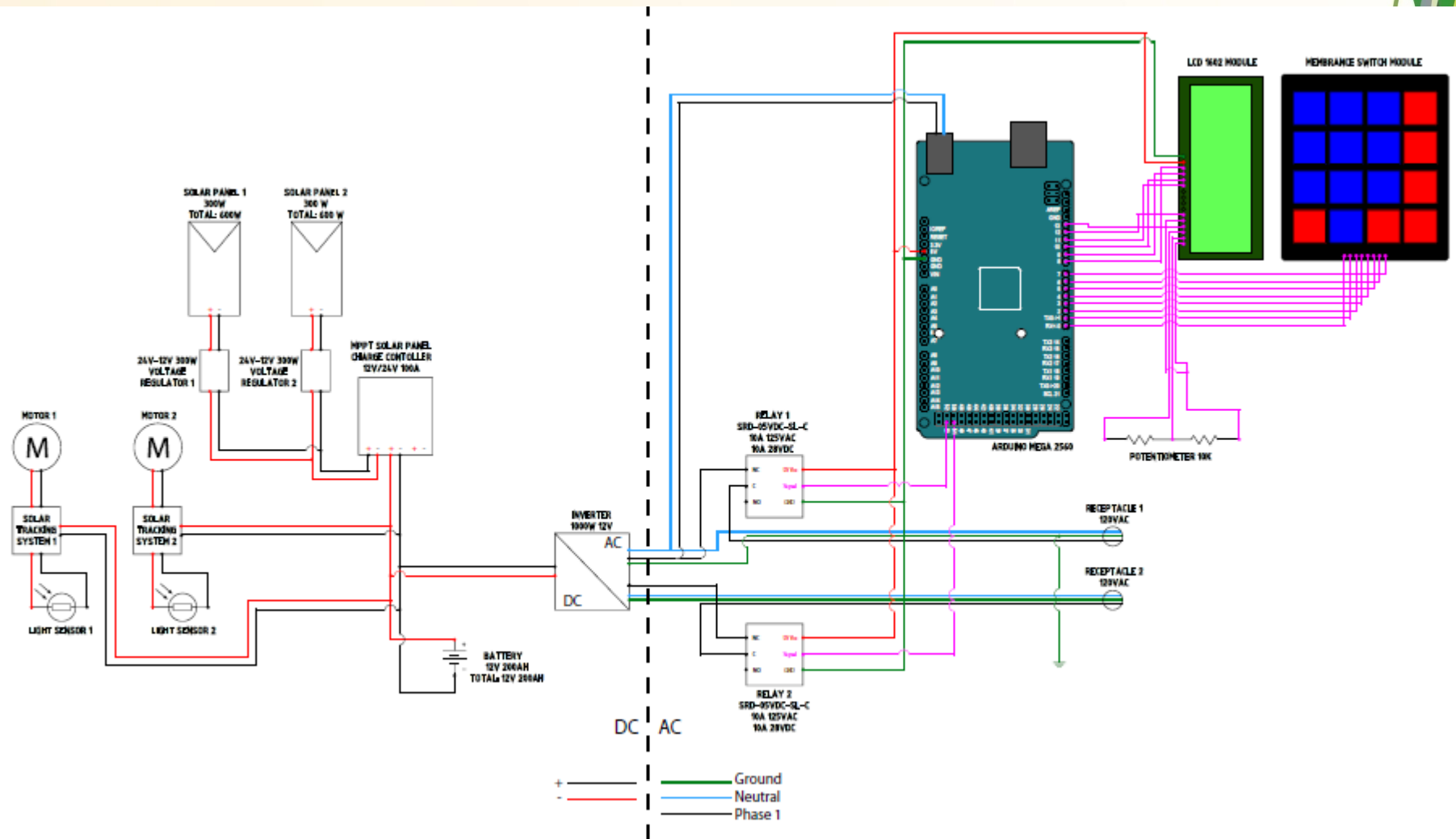
void loop() {
  // put your main code here, to run repeatedly:
  int sensorValue = 0;
  sensorValue = analogRead(A0);
  float voltaje = sensorValue * (0.0048)*(2.70)*(11.11);

  lcd.print(voltaje);
  lcd.setCursor(6,0);
  lcd.println(" ");
  lcd.setCursor(0,0);
  delay(5000);
  Serial.print("  ");
  Serial.println(voltaje);

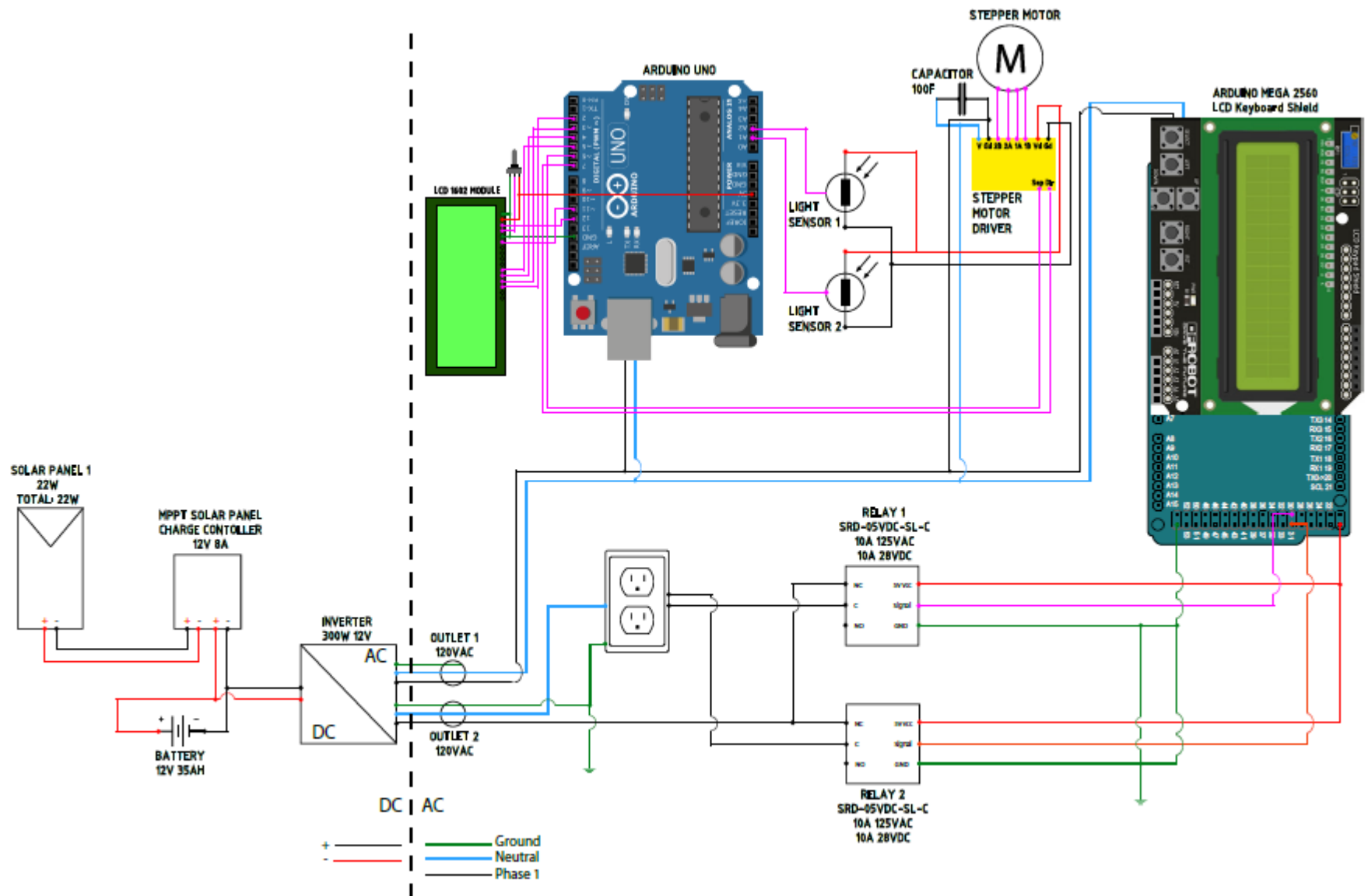
  delay(5000);
```

Figure 17: Arduino code

Proposal Circuit Design



Prototype Circuit Design



Circuit simulation

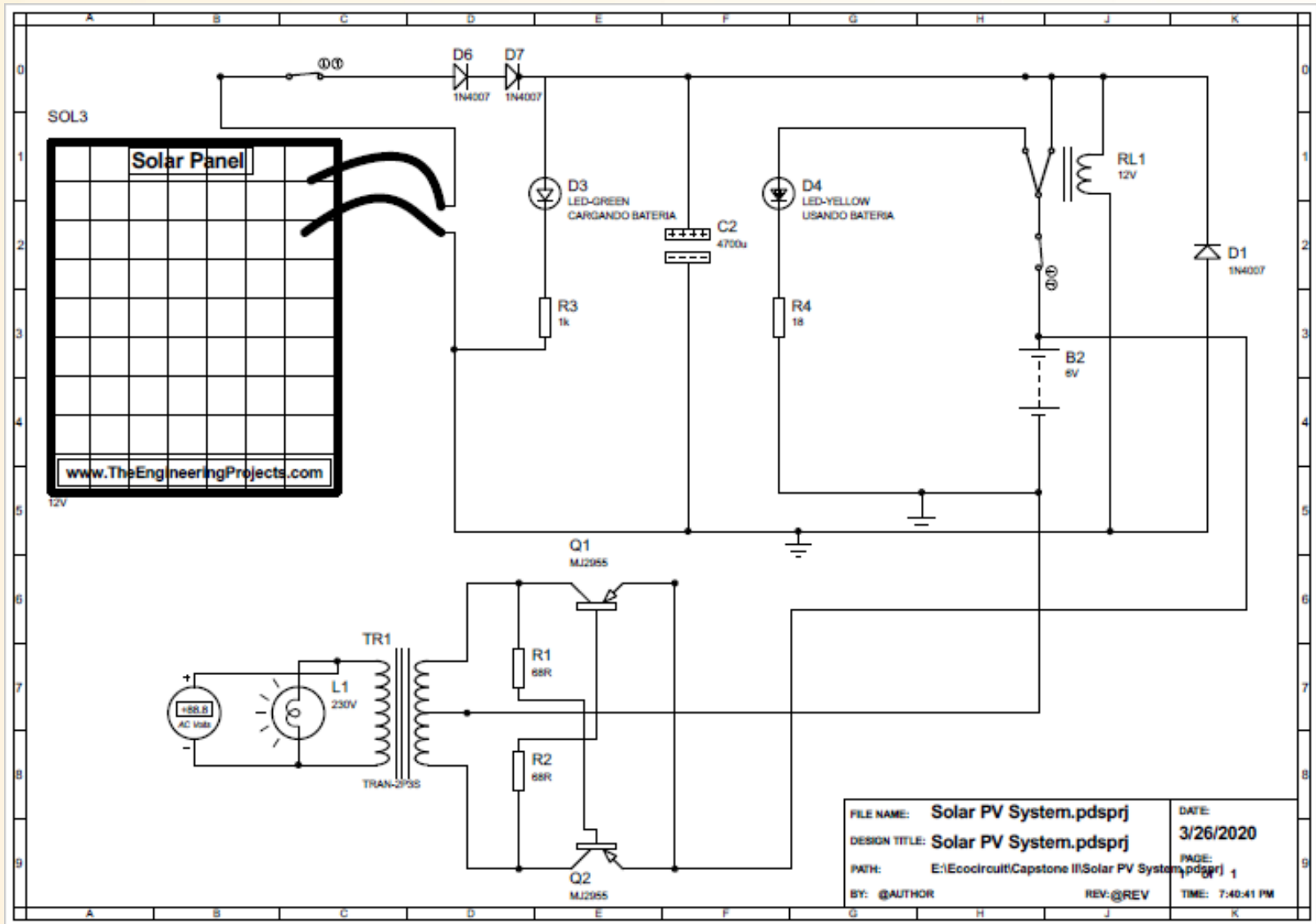


Figure 18: Circuit simulation of the project

Solar Panel Analysis

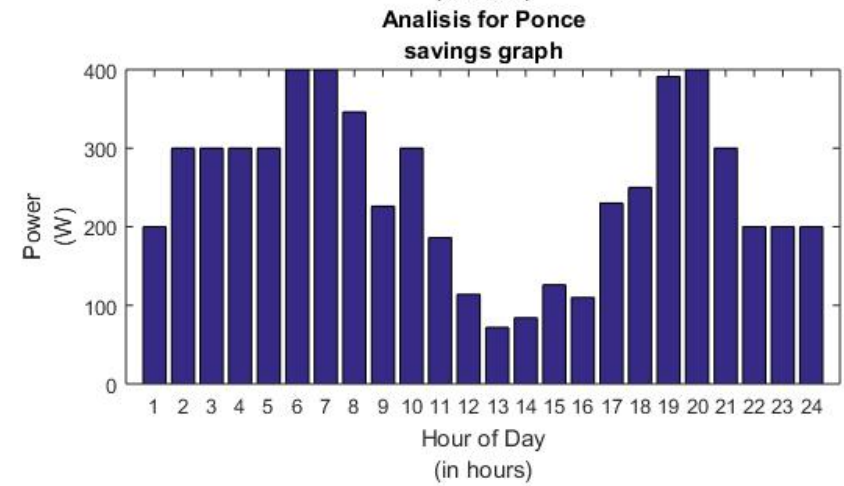
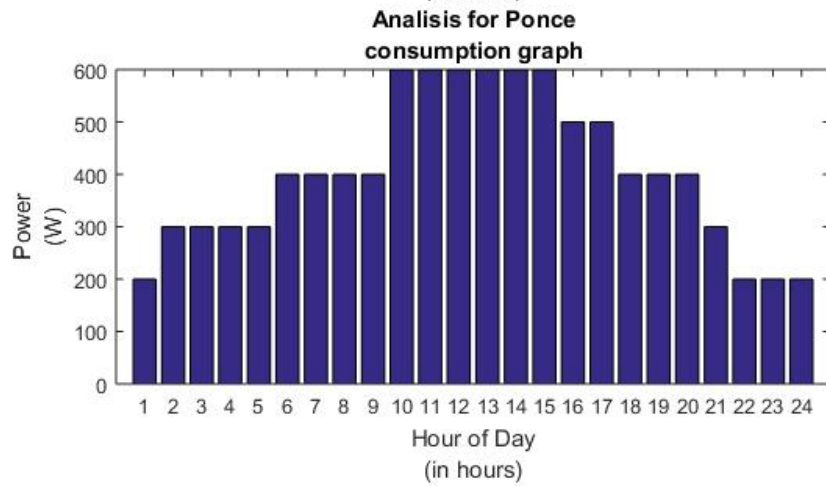
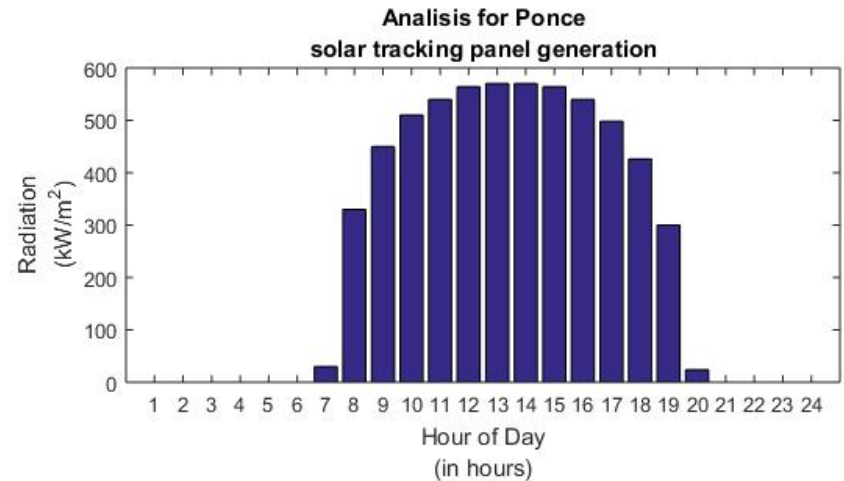
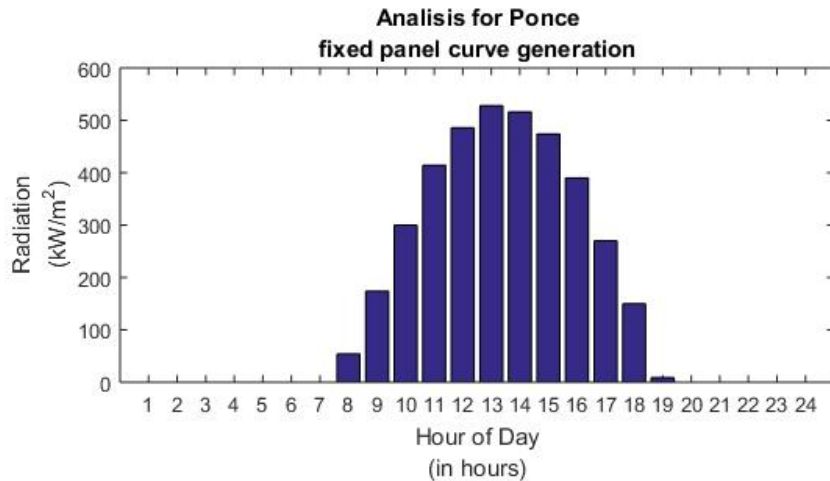


Figure 19: graphs of solar panels analysis

Proposal Calculations

Table 9: components calculations

PV System Calculator (Proposal)

A. Components					
Qty.	Components	Watts	Hour/Day	Wh/dia	20% Security Margin
1	Small fridge	27	24	648	777.6
1	Router #1 (HT2000W)	75	10	750	900
0	Component AC			0	0
2	Solar Tracking System (Max Load)	36	0.5	36	43.2
2	Solar Tracking System (No Load)	10	1	20	24
1	Fiber Optic Media Converter	3	10	30	36
1	Microcontroller	0.25	3	0.75	0.9
1	LCD Display	0.01	3	0.03	0.036
1	Keyboard	0.01	3	0.03	0.036
1	USB Port #1 (Cellphone)	6	6	36	43.2
0				0.0	0.0
		157.3		1520.8	1825.0

Table 10: design specs calculations

B. Design Specification	
Minimum inverter wattage	722 W
Minimum battery AH reserve	187 Ah
Minimum solar panel modules	1.4 Panels
	1.0 Serie
	1.4 Parallel
Minimum charge controller current	45.4 A

Prototype Calculations

Table 11: components calculations

PV System Calculator (Prototype)

A. Components					
Qty.	Components	Watts	Hour/Day	Wh/dia	20% Security Margin
1	Router	7.5	2	15	18
1	Modem	6	2	12	14.4
0				0	0
1	Arduino Mega	0.27	2	0.54	0.648
1	Arduino UNO	0.29	2	0.58	0.696
2	Relays	0.25	2	1	1.2
1	LCD Keyboard Shield	0.25	2	0.5	0.6
1	Stepper motor	24	0.25	6	7.2
0				0	0
1	Celular	6	2	12	14.4
0				0.0	0.0
Table 12: design specifications calculation		44.6		47.6	57.1

B. Design Specification

Minimum inverter wattage	46 W
Minimum battery AH reserve	8 Ah
Minimum solar panel modules	0.8 Panels
	1.0 Serie
	0.8 Parallel
Minimum charge controller current	1.7 A

Current Calculations

Table 13: currents comparison

Current Calculated vs Measured

Components	Watts (W)	Voltage (V)	Current (A)	
			Calculated	Measured
Arduino Mega	0.27	9.0	0.030	0.09
Arduino UNO	0.29	9.0	0.032	0.05
Relay x2	0.35	5.0	0.07	0.08
Stepper Motor	24.00	12.0	2.00	1.65
ARRIS TG862A Modem	6.00	12.0	0.50	0.48
Linksys E1000 Router	7.50	12.0	0.63	0.50

Systems comparison



Figure 20: Communication & Charge Portable System

- Wi-Fi
- Battery
- AC Outlet
- Satellite Antenna



Figure 21: Solar charge station
by © 2015 SUNSUPPLY

- Charge 6 phones
- Portable
- Solar



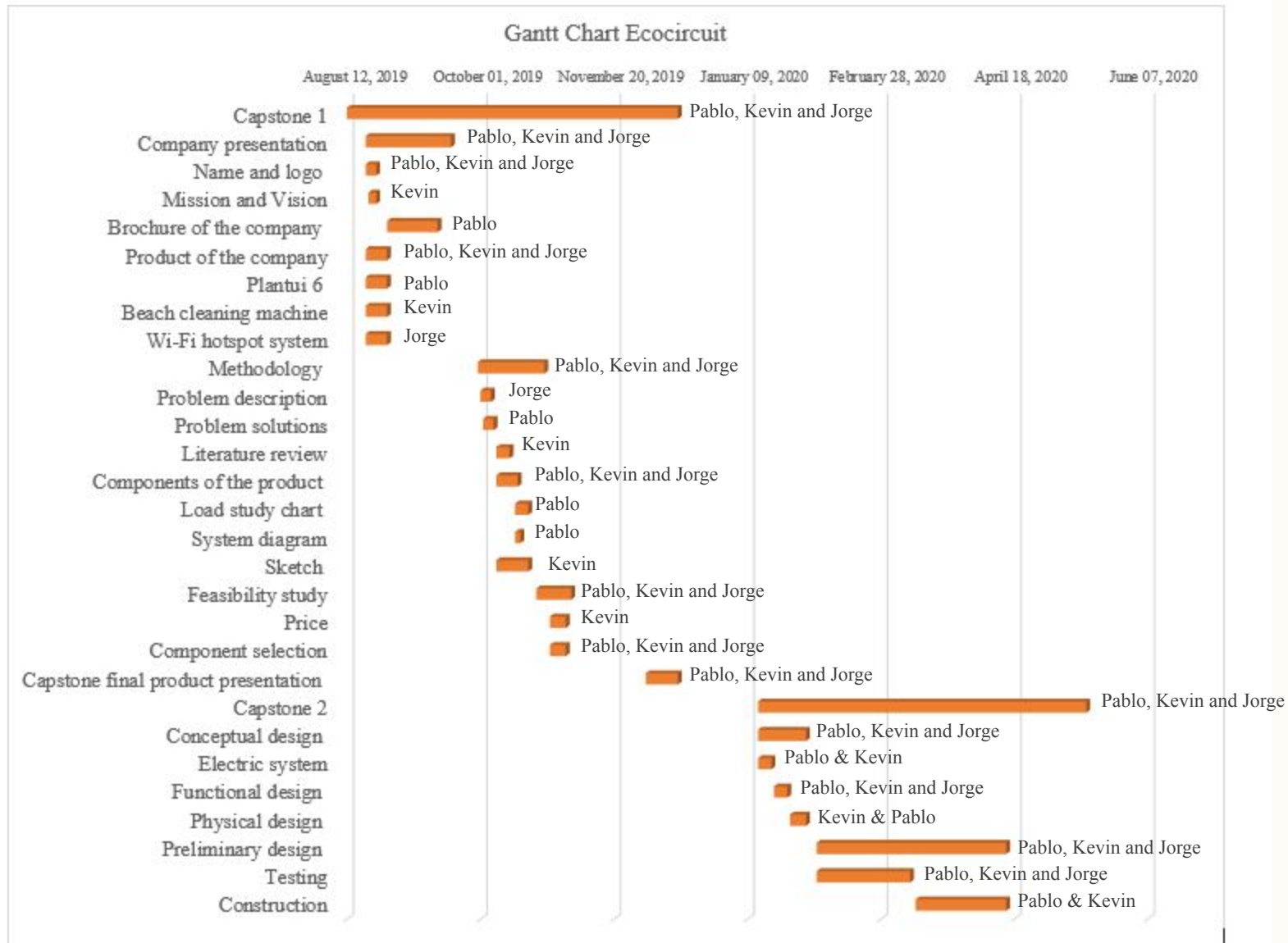
Figure 22: Ecocircuit
smaller prototype

- Wi-Fi
- Battery
- AC Outlet
- Satellite Antenna

Economic Analysis

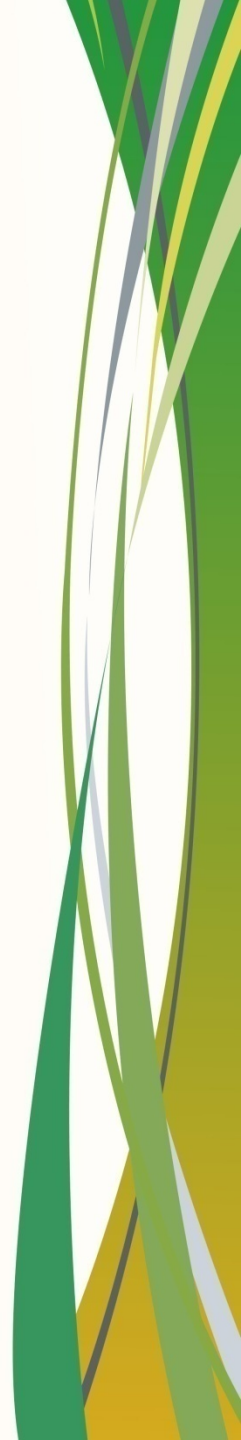
- Dairy production – dairy consumption = result/1000 * 0.215 (value of kWh in PR)
- $5930 - 1520 = 4410 / 1000 * 0.215 = 0.95 * 360 = 341.33$
- Annual economic analysis is: \$341.33
- Monthly economic analysis is: \$28.44

Gantt Chart



Conclusion

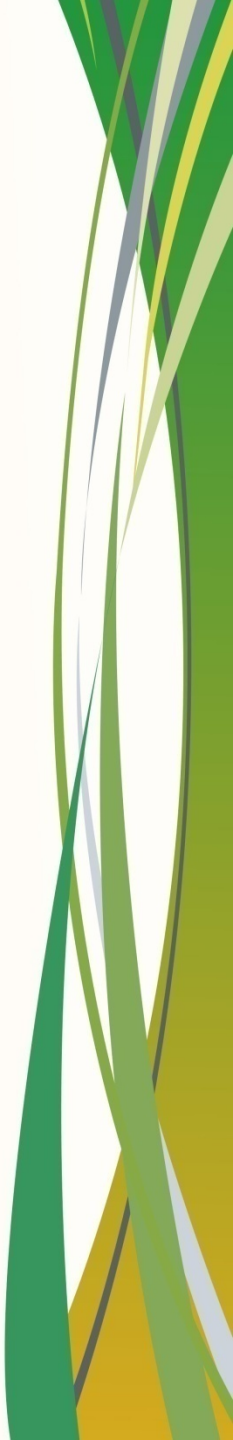
- After making the prototype at the system scale, we were able to demonstrate that our solar system has a correct operation.
- We were able to establish that our system can power a communication system.
- We also demonstrate the operation of the microcontrollers since they were programmed to control the system, create a functional solar tracking system and a system to measure the voltage of our battery.



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Any question?



Annex

HughesNet HT2000W .98m 3-watt Ka-Band Gen5 Jupiter 2 High-Speed Satellite Internet System

Power supply type	Description	Electrical requirements
AC/DC, Ravel 75 W P/N 1504941-0001	HT2000W satellite router with 1 W radio only	Input line voltage: 100 - 130 V, 1.5 A maximum Input line frequency: 60 Hz AC Rated output power: 75 W



Annex

Parts	Solar panel to charge controller	Charge controller to battery	Turbine to Battery	Charge controller to inverter
Types of cable AWG	3	3	4	4
Voltage drop	0.83	0.83	0.85	0.85
Amp.	75	75	60	60